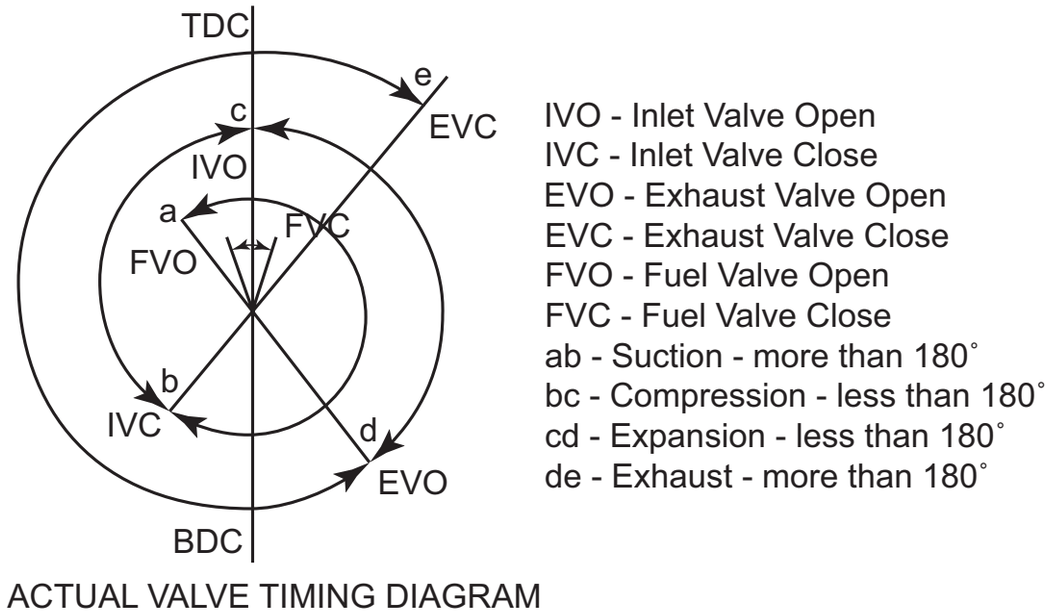


Fig 5.7(a) Valve Timing Diagram of Four Stroke Diesel Engine

TDC. If the inlet valve is allowed to close at BDC, the cylinder would receive less amount of air than its capacity. To avoid this, the inlet valve is kept open for 25°to 45°after the BDC, during the compression stroke. This will ensure the full induction of fresh charge in cylinder during suction stroke.

After the inlet valve closed in the compression stroke, the air is compressed by the upward motion of the piston. This will increase the pressure and temperature inside the cylinder.

With both intake and exhaust valve are in closed condition, the fuel injector will inject the diesel fuel at high pressure at 5° to 10° before TDC. The temperature of the compressed air at the end of compression stroke is sufficient enough to initialise the finely automised diesel fuel for combustion. The air fuel mixture will be burned in the combustion chamber. Thus the chemical energy of the fuel is converted into heat energy. Based on the speed of the engine, the fuel is injected even 25° before TDC.

Exhaust valve timing:

The exhaust valve opens at 30° to 60° before the completion of power stroke. Due to this, the gases have an outlet for expansion, which removes the greater part of the burnt gases. The valves remain open during open during the exhaust stroke and it is closed at 20° after TDC, during the intake stroke of the next cycle.

Scavenging

The inlet valve will be open 10° to 30° before TDC during exhaust stroke and at the same time the exhaust valve is already in the open condition. This will make the fresh charge to push out the burnt out gases from the cylinder. Scavenging will be made till 20° after TDC.

AM_Unit_05.indd 109 12/20/2019 5:26:41 PM